

L11: Kuramoto Model on Complex Networks

What happens when the N oscillators are coupled through a complex network? How does the topology of that network affect whether the oscillators get synchronized, and the critical coupling strength?

This problem has received significant attention in the last few years – and there are limited results but only for special cases and only under various assumptions.

First, in the case of an undirected network with adjacency matrix $\mathbf{A}$, the Kuramoto model can be written as follows,

$$\frac{d\phi_i}{dt} = \omega_i + K \sum_{j=1}^N A_{i,j} \sin(\phi_j - \phi_i)$$

A common approximation (referred to as “degree block assumption” – we have also seen it in Lesson-9) is to ignore any degree correlations in the network, and substitute $A_{i,j}$ with its expected value according to the configuration model:

$$A_{i,j} \approx k_i \frac{k_j}{2m} = \frac{k_i k_j}{N\bar{k}}$$

where $\bar{k} = 2m/N$ is the average node degree, and m is the number of edges.

Under this approximation, the Kuramoto model can be written as:

$$\frac{d\phi_i}{dt} = \omega_i + \frac{K}{\bar{k}} \sum_{j=1}^N \frac{k_i k_j}{N} \sin(\phi_j - \phi_i)$$

Note that as $N \rightarrow \infty$, the summation remains finite – and non-zero.

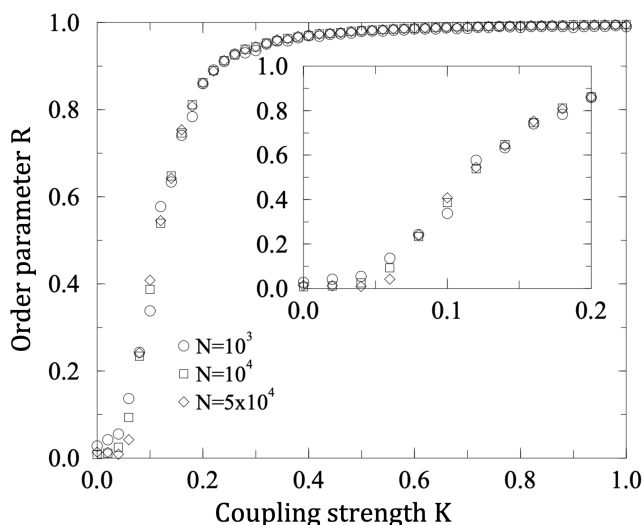
Under this approximation, a necessary condition for the onset of synchronization is that the coupling strength $\frac{K}{\bar{k}}$ is larger than the following critical threshold:

$$K_c = \frac{2}{\pi\Omega} \frac{\bar{k}}{\bar{k}^2}$$

where $\bar{k}^2$ is the second moment of the degree distribution.

This formula predicts a very interesting result: *for power-law networks with degree exponent $2 < \gamma \leq 3$, the network will always get synchronized, even for a very small K , because the second moment $\bar{k}^2$ diverges.*

Image Source: “[Synchronization in complex networks](https://core.ac.uk/download/pdf/71321767.pdf)” by Arenas et al.



The visualization shows simulation results of heterogeneous oscillators on power-law networks with degree exponent $\gamma = 3$ (the networks are constructed with the Preferential Attachment model). The natural frequencies of the oscillators vary uniformly in the interval $[-1/2, 1/2]$. The onset of synchronization is not exactly at 0 – but close to 0.05. The small deviation from the theoretically predicted result does not seem to be a consequence of finite N because the critical threshold remains close to 0.05 even as N increases.

We have seen this ratio of the first two moments of the degree distribution several times earlier in the course, including in the friendship paradox, the epidemic threshold, and the critical threshold for random failures. When the second moment diverges, we saw in Lesson-9 that the epidemic threshold goes to 0, and

earlier in this Lesson that the critical threshold for random failures goes to 1. Something similar happens here: networks with major hubs get easily synchronized, even with very minor coupling strength.